



Climate Shocks and the Financial Health of American Households and Hospitals

Three Papers: Incidence, Persistence, and Inequality

Ahwaz Akhtar · George Washington University · June 2026

Overview

Three papers on the financial consequences of climate shocks across the health system — households and the hospitals that serve them — on one U.S. panel

Three papers

- **Incidence** — do climate shocks raise the financial burden of health, for households and hospitals?
- **Persistence** — do the effects reverse, scar, or compound?
- **Inequality** — does structural vulnerability amplify them?

Empirical approach

- Two-way fixed effects with distributed lags; local projections for dynamics
- Natural-experiment and Callaway–Sant’Anna difference-in-differences against never-exposed counties
- Onset/persist/exit symmetry; cumulative-dose; hazard $\times$ exposure $\times$ vulnerability index
- Estimates mirrored at state and county level

Three Papers: Incidence, Persistence, Inequality

Incidence (how much) → Persistence (how long) → Inequality (who bears it)

Paper	Question	Design	Headline
1. Deferred Costs (Incidence)	Do climate shocks raise the financial burden of health?	Distributed-lag FE; never-exposed DiD; local projections	Cold (lag 1) and drought (lag 2) raise medical debt and premiums via an income channel
2. Scarring & Compounding (Persistence)	Do the costs reverse, scar, or compound?	Onset/persist/exit symmetry; cumulative-dose; Callaway–Sant’Anna	Drought debt scars ; cold employment compounds ; heat saturates
3. Unequal Weather (Inequality)	Who bears the cost?	Hazard × exposure × vulnerability (CDC SVI); EJ interactions	Harm to income, jobs, premiums is amplified where vulnerability is highest

Econometric Robustness and Methodological Additions

Five analyses extend the core specification; each addresses a point raised in prior review

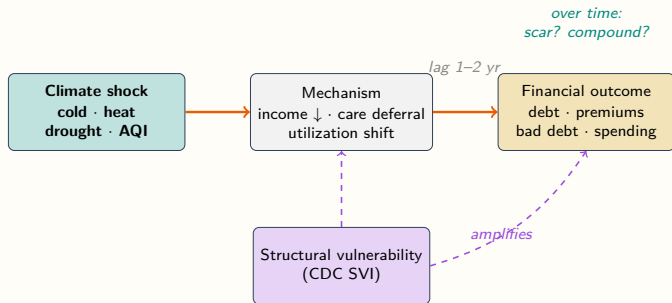
Analysis	Result	Paper
Random effects (Hausman)	Rejected for all 11 outcomes ($p < 10^{-9}$); fixed effects retained	1
Post-exit / transition dynamics	Onset/persist/exit local projections; drought income partially scars	1, 2
Natural-experiment DiD	2012 drought 2×2 and Callaway–Sant’Anna vs. never-exposed counties	1, 2
Propagation pathways	18 references and descriptive figures, one set per channel	all
Humidity (PRISM t_{dmean})	Acquired and aggregated to state and county; headline findings stable	2

Conceptual Model and Hypotheses

What the propagation-pathway literature predicts —
stated *ex ante*, before consulting our estimates

Conceptual Model: From Climate Shock to Health-Financial Outcome

A lagged causal chain, moderated by vulnerability — the three papers interrogate three parts of it



Three questions, three papers

- **Paper 1 — the chain:** sign and lag of shock → outcome (orange arrows)
- **Paper 2 — over time:** does the effect reverse, scar, or compound? (teal)
- **Paper 3 — moderation:** does **vulnerability** amplify it? (purple)

One model; each paper isolates a different margin of the same chain.

Proposed Mechanism Pathways and Supporting Evidence

Each headline channel is anchored to peer-reviewed evidence on its direction and time-scale

Channel	Mechanism	Key references
Heat → delayed care	Acute heat defers non-urgent ambulatory visits; substitution into later high-acuity ED/inpatient raises cost with a lag	Sun et al. '21; White '17; Karlsson–Ziebarth '18
Cold → utilization shift	Respiratory/cardiac demand rises on impact; heating costs crowd out health spending (“heat-or-eat”), raising missed payments	Deschênes–Moretti '09; Gasparrini et al. '15; Andrews et al. '17
Drought → income → debt	Farm and adjacent income falls; local consumption and ability-to-pay erode over 1–3 years — an income, not biological, channel	Hornbeck '12; Burke–Hsiang–Miguel '15; Currie–Greenstone–Meckel '17
AQI → respiratory/cardiac	PM _{2.5} /ozone raise acute respiratory and cardiovascular events, feeding ED/inpatient utilization and cost	Deryugina et al. '19; Schlenker–Walker '16; Currie–Walker '11

18 references across the four channels; full set in the propagation-pathways review.

Hypotheses: What Existing Literature Predicts

Direction and time-scale committed *ex ante* from the propagation-pathway literature — the “where should we be surprised?” lens

Pathway	Predicted effect	Time-scale
Drought → income → debt	Debt ↑ (via income ↓)	lag 1–2 yr
Cold → utilization shift	Debt & premiums ↑	contemp.–lag 1
Heat → delayed care	Spending ↑ (deferred)	lag 1–2 yr
AQI → respiratory/cardiac	Utilization ↑	within-year

Structural hypotheses

- H1 Incidence:** shocks raise the financial burden of health with a 1–2 yr lag (income + utilization channels)
- H2 Persistence:** drought *scars* (Hornbeck long-run); cold expected *acute* (Deschênes-Morette) — test if cumulative exposure compounds
- H3 Inequality:** structural vulnerability *amplifies* harm (EJ / Lancet Countdown)

Ex-ante vs. realized

- Drought lag-2, cold lag-1 → debt: **as predicted**
- Long-run cold *compounding*: **stronger than predicted**
- Humidity → debt: no prior

Paper 1 — Deferred Costs

Incidence: climate shocks raise the financial burden of health — on households and hospitals,
with a one- to two-year lag

Paper 1: Prior Evidence from the Panel

Two-way fixed effects with distributed lags, state and county; selected significant coefficients

Coefficient	Estimate	
<i>State</i>		
Employer premium: Severe Drought (lag 0/1)	+\$86 / +\$65	**
Debt share: Cold Shock (lag 1)	+0.0135	**
Total PC health exp: Cold Shock ($h=0$)	+\$205	**
Medicaid PE: Severe Drought (lag 1)	-\$525	***
<i>County</i>		
Benchmark Silver: High HDD ($h=0$)	+\$22	***
Debt median (PW): High HDD ($h=0$)	+\$31	**
Hospital bad debt: High HDD ($h=0$)	+\$5.3	***
Med HH income: Z_Temp ($h=0$, heat)	-\$88	***
PCPI: PDSI (lag 2)	-\$113	**

What this establishes

- Cold and drought raise medical debt and premiums with a **1–2 year lag**
- Transmission runs through **household income**, not systemic spending
- State and county estimates agree in direction

What it cannot settle

- Within-unit FE identifies an *association*
- Recurring shocks confound a clean before/after comparison
- Motivates a sharp natural experiment

** $p < 0.05$, *** $p < 0.01$; dollar outcomes in \$2023.

The 2012 Midwest Drought: Context for a Natural Experiment

One of the most severe and extensive U.S. droughts in decades — a sharp, plausibly exogenous shock

The event

- Severe drought across the Corn Belt and Great Plains; widespread crop losses and federal disaster declarations
- **139 counties** record their *first* extreme-drought event ($\text{PDSI} \leq -4$) in 2012 — a 4.4% onset rate
- A documented event with a defined geography and timing

Difference-in-differences design

$$Y_{it} = \alpha_i + \gamma_t + \tau (\text{Treated}_i \times \text{Post}_t) + \varepsilon_{it}$$

- **Treated:** first-ever extreme drought = 2012 (139 counties)
- **Control: never-exposed** counties, 2011–2023 (2,534)
- Cluster: state; event-study reference year $t = -1$

A sharp event against never-exposed controls closes the recurring-shock loophole that within-county lags cannot.

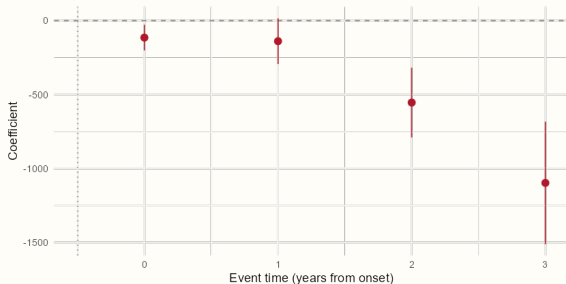
Paper 1: What the Natural Experiment Adds

Treated vs. never-exposed counties: income and employment fall, accumulating over the event window

2012 Drought → Civilian Employed

Event Study (DiD): Drought_2012 → Civilian_Employed

Reference: $t = -1$; cluster = State



Near-zero pre-period; post-event loss accumulates (~ -100 at $t=1$ to $\sim -1,100$ at $t=3$).

2×2 ATT	Estimate	p
PCPI Real	−\$1,311	0.027
Civilian Employed	−2,053	0.0001
Med HH Income Real	−\$992	0.24
Medical Debt Share	−0.0062	0.49

Prior + experiment

- Panel: drought/cold → debt and premiums (lagged, income channel)
- Experiment: **causal** income and employment loss vs. never-exposed
- Debt is null in the 2×2 (averages over post years; LP peaks at $h=1-2$)

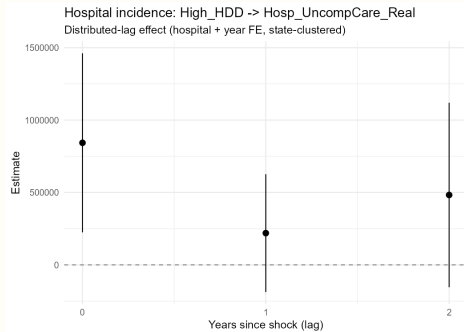
Paper 1, Supply Side: Climate Shocks Push Unpaid Bills Onto Hospitals

The same shocks that raise household medical debt land on provider balance sheets — hospital (CCN) $\times$ year panel (NASHP HCT), 59,896 hospital-years, hospital + year FE

Where the household side showed debt, the hospital side shows uncompensated care

Shock $\rightarrow$	Effect	p
uncompensated care		
Cold (HDD) $\rightarrow$ \$ uncomp. care	+\$1.55M	0.022
Cold (HDD) $\rightarrow$ % of NPR	+0.53 pp	0.097
Heat (CDD) $\rightarrow$ % of NPR	+1.22 pp	0.063
Drought $\rightarrow$ % of NPR (measured)	-1.68 pp	0.032

- Temperature shocks push *unpaid bills* onto hospital ledgers — the provider mirror of Paper 1's household debt channel
- Average operating / net margins *not* moved: the strain is absorbed through uncompensated care, not reported margin



Cold-shock distributed-lag effect on uncompensated care (\$2023); drought's negative sign is a measurement caveat.

Paper 2 — Scarring and Compounding

Persistence: do the costs reverse, scar, or compound?

Onset/persist/exit symmetry, cumulative-dose, chronic-exposure cohorts

Paper 2: From Levels to Dynamics

Level effects show shocks raise costs; post-exit local projections motivate the persistence question

What we knew

- Cold and drought raise debt and premiums (Paper 1; levels and lags)
- Post-exit LP: drought income *partially scars*
— +\$1,044 PCPI at $h=2$ ($p = 0.0002$)
- Post-exit LP: cold-shock exit brings *immediate* hospital-cost relief (-3.1 at $h=0$)
— no within-county scarring

Open questions

- Is the cost of *entering* a shock mirrored by the benefit of *leaving* it? (symmetry)
- Does *cumulative* exposure compound, or does the effect saturate?
- Does the answer differ by hazard?

The level/lag specification cannot separate a reversible shock from a scarring or compounding one.

Paper 2: Two Designs for Persistence

Onset/persist/exit symmetry, and a cumulative-shock-years dose-response

Transition symmetry

- Decompose each shock into **onset** ($0 \rightarrow 1$), **persist** ($1 \rightarrow 1$), **exit** ($1 \rightarrow 0$) vs. never-transitioned
- Estimate the three jointly by local projection, $h=0 \dots 3$
- Test $\beta_{\text{onset}} + \beta_{\text{exit}} = 0$: a nonzero sum is hysteresis (scarring or over-relief)

Cumulative-dose response

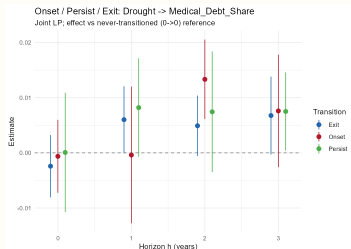
- Count **cumulative shock-years** per county (monotone, non-resetting)
- Regress outcomes on linear, quadratic, and binned dose (1–3/4–6/7–9/10+)
- Compare the marginal effect of the **10th** shock-year to the **1st**

Both estimated with county + year fixed effects, state-clustered; cross-validated against a Callaway–Sant’Anna design.

Paper 2: Three Hazards, Three Persistence Profiles

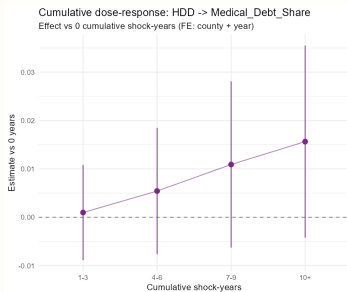
Drought scars (onset/exit asymmetry); cold compounds and heat saturates (cumulative dose)

Drought — scars



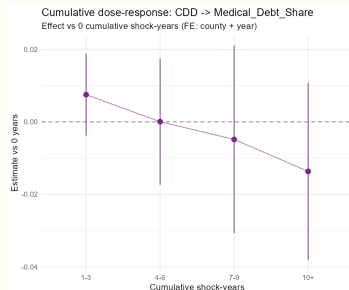
Onset and exit both raise debt at $h=2$ — not undone on exit.

Cold — compounds



Debt share rises with each cumulative cold-year.

Heat — saturates



Marginal debt effect flattens and reverses by year 10.

Headline: persistence is shock-specific — drought onset/exit asymmetry +0.018 ($p=0.0015$); cumulative cold raises debt to +1.5 pp at 10+ years (and compounds employment loss, -5,668 jobs);

Paper 2, Supply Side: Drought Scars Hospital Balance Sheets

The county-level “drought scars, temperature reverses” dynamic reappears on hospital finances — same onset/exit symmetry and cumulative-dose tools, hospital + year FE

Onset + exit asymmetry ($h = 0$)

Drought → outcome	Asymmetry	p
Uncompensated care, %NPR	−0.013	0.012
Operating margin	+0.027	0.006

- Drought onset and exit *do not cancel* — hysteresis on both hospital outcomes (**scarring**), echoing the county debt scar in Paper 2
- Cold / heat shocks are *symmetric* (reversible), as on the household side

Cumulative exposure

- Operating-margin dose-response is mildly *positive* (10+ vs 1–3 yrs +0.018, $p = 0.033$)
- Read as **survivorship / adaptation** in an unbalanced panel — *not* margin erosion
- The supply-side persistence story is drought *hysteresis*, not dose-driven decline

Same “drought scars, temperature reverses” pattern the county delta analysis found in Paper 2, now on hospital balance sheets.

Paper 3 — Unequal Weather

Inequality: who bears the cost?

A hazard $\times$ exposure $\times$ vulnerability index (CDC SVI)

Paper 3: From Average Effects to Their Distribution

Papers 1–2 estimate population averages; who bears the cost is a separate question

What the averages miss

- Papers 1–2 estimate *population-average* effects of each shock
- They do not identify **which communities** absorb the cost
- A given shock may be far more costly where households are already fragile

Hypothesis (environmental justice)

- Structural vulnerability **amplifies** climate harm
- Grounded in the environmental-justice and Lancet Countdown vulnerability-indicator literature
- Prediction: the same shock costs more at higher vulnerability

Paper 3: A Hazard $\times$ Exposure $\times$ Vulnerability Index

Climate hazard interacted with the CDC Social Vulnerability Index; estimates mirrored at state and county

Construction

- Combine **hazard** (shock) $\times$ **exposure** (population) $\times$ **vulnerability** (CDC SVI percentile)
- Composite scalar index, standardized
- Lancet-style **person-years** of extreme-temperature exposure

Interaction design

- Estimate **Shock** $\times$ **SVI** in the fixed-effects model
- Recover the marginal shock effect at **low (p25)** vs. **high (p75)** vulnerability
- Mirror every estimate at state and county level

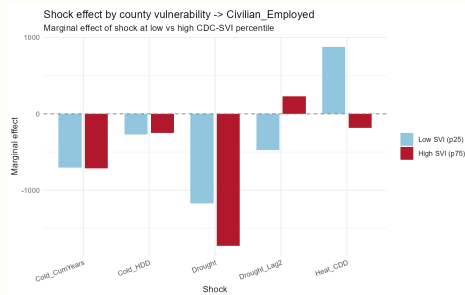
Paper 3: Structural Vulnerability Amplifies Climate Harm

Shock $\times$ CDC Social Vulnerability Index; marginal effect at low (p25) vs. high (p75) vulnerability

Shock $\rightarrow$ outcome	Low SVI	High SVI	p_{int}
Heat $\rightarrow$ employment	+878	-184	0.001
Cold $\rightarrow$ PCPI	-\$56	-\$472	0.056
Drought (lag 2) $\rightarrow$ premium	-\$54	+\$16	0.001

Separately, the *Composite Heat Exposure Index* — a single standardized index, with no SVI split — lowers median income by **-\$435/SD** ($p = 0.0002$): the index predicts harm on average.

- In vulnerable counties heat **costs** jobs; cold's income hit is $\sim 8\times$ larger; drought pushes premiums *up*
- Lancet-style exposure: $\sim 70\text{--}105\text{M}$ heat person-years/yr ($3\text{--}5\times$ cold)



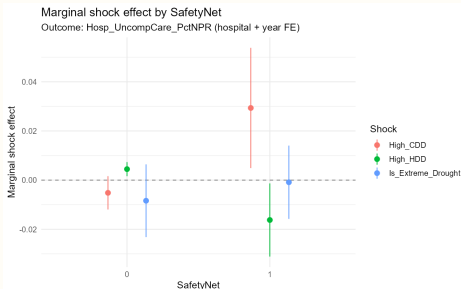
Heat's employment effect at low vs. high vulnerability — it flips from gain to loss.

Paper 3, Supply Side: Strain Concentrates in Safety-Net Hospitals

The provider analogue of the SVI amplification: where high-SVI *counties* bore more household harm, high-Medicaid *safety-net hospitals* bear more heat-driven uncompensated care — Shock $\times$ moderator, hospital + year FE

Shock $\times$ moderator $\rightarrow$ uncomp. %NPR	p_{int}	Concentrates?
Heat (CDD) $\times$ Safety-net	0.021	Yes
Cold (HDD) $\times$ Safety-net	0.011	reversed
Cold (HDD) $\times$ Medicaid exp.	< 0.001	expansion
Drought $\times$ Medicaid exp.	< 0.001	expansion

- **Heat-driven** uncompensated care concentrates in *safety-net* hospitals — those least able to absorb it
- Amplification is shock- and outcome-specific (as on the household side), *not* blanket
- Medicaid-expansion interactions are strong but point (surprisingly) to *expansion* states



Marginal shock effect on uncompensated care at non-safety-net vs. safety-net hospitals.

Credit-Bureau Medical Debt: Measurement Issues

Income, employment, and premium results are robust; the medical-debt outcome is sensitive to how debt is recorded

Pattern

- Debt response concentrates in *less*-vulnerable counties
- Sign **reverses** between county and state aggregation

Interpretation

- Credit-bureau debt requires insurance, billed encounters, a credit file
- Higher-uninsurance areas record less *measured* debt
- Consistent with a recording artifact, not absence of harm

Cross-level evidence

- Estimates mirrored: humidity → county, SVI → state, demographics → state
- Income, employment, and premium effects **replicate** across aggregation
- Only medical debt is aggregation-sensitive

The real-economy outcomes carry the inequality result; the debt outcome is reported with the recording caveat.

Humidity (PRISM td_{mean}) Does Not Overturn the Findings

Mean dew point added as a control at state and county level; headline coefficients are stable

Procedure

- PRISM annual td_{mean} (mean dew point), 2009–2025
- Area-weighted to **state and county** polygons
- Added as a control; headlines compared with vs. without humidity on a constant sample

Result

- State: cold (lag 1) $\rightarrow$ medical debt stable, 0.0136 $\rightarrow$ 0.0137
- County: cold $\rightarrow$ hospital bad debt and premiums essentially unchanged
- Humidity itself predicts medical debt (positive)

The drought lag-2 and cold lag-1 results are not driven by humidity confounding.

Summary

Findings across the three papers, and open issues

Findings

- **Incidence:** cold (lag 1) and drought (lag 2) raise medical debt and premiums, via a household income channel
- **Persistence:** drought debt scars; cold employment compounds with cumulative exposure; heat saturates
- **Inequality:** income, employment, and premium harm is amplified where structural vulnerability is highest
- **Supply side (Ch. 2):** each paper has a hospital companion — temperature shocks raise uncompensated care,

Open issues

- Credit-bureau medical debt is measurement-sensitive
- Cumulative-CDD difference-in-differences carries a north/south regional confound — suggestive only
- Multiple testing: headline cells survive Bonferroni correction within shock

Findings are consistent across state and county aggregation throughout.

Thank you.

Discussion and questions